

Beaconhouse National University

BS Computer Science

Final Year Project

DAILYKHATA

Project Governance Project Scope

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Revision History

Historical source versions remain unchanged in their original locations. This replacement consolidates scope, terminology and evidence provenance under the recorded decisions. Version 2.0, 23 September 2026, prepared for the DailyKhata Project Team.

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1 Authority and purpose

This is the replacement scope authority for DailyKhata D1 v2.0 and D2 v2.0. It consolidates the project owner instructions dated 23 September 2026 and the existing Scope Baseline v1.0. The original baseline is preserved. A recorded decision is required before a later deliverable changes a project fact, scope boundary or priority. Supervisor endorsement of these replacements has not been recorded. [G1, U1]

DailyKhata is a personal expense management application primarily designed for Pakistani users who frequently deal with paper receipts, cash transactions and informal expense tracking. It reduces the effort required to maintain personal or household expense records through receipt digitisation, OCR, quick cash expense capture, manual expense entry, automatic categorisation, user confirmation and correction, a structured Chart of Accounts, transaction history, spending summaries, item-level price history, semantic product classification and an AI Financial Assistant.

2 Users and operating boundary

Primary users are homemakers and household budget managers, university students and salaried professionals. Small shopkeepers are secondary direct users for petty cash, supplier expenses, receipt capture and expense summaries. Family members and advisers are secondary beneficiaries of information shared by the account holder.

The baseline is a single-account expense-management system. Household expenses can be recorded by the account holder and reviewed together. CSV and/or PDF export is Should Have; a dedicated exportable household summary is Could Have. Shared household accounts, invitations, household permissions and collaborative editing are not current requirements.

2.1 Current exclusions

Direct bank integration; automatic debit-card or credit-card synchronisation; Easypaisa and JazzCash transaction integration; open banking; payment processing; money transfers; inventory management; point-of-sale functionality; payroll; enterprise ERP; complete SME accounting; tax filing; investment management; credit scoring; fraud-detection infrastructure; commercial banking functionality; enterprise multi-tenant organisation management; banking-grade regulatory certification; production infrastructure for millions of concurrent users; arbitrary third-party financial integrations.

2.2 Future scope

Bank transaction importing; wallet integrations; shared household accounts; collaborative budgeting; business accounting features; inventory; POS; recurring expenses; budgeting goals; predictive analytics; anomaly detection; advanced financial recommendations; deeper Urdu support; additional languages; multi-currency support; retailer integrations; e-receipt ingestion.

A feature can be excluded from the current project and also be named as potential future work. Future listing is not a delivery promise. Recording an expense paid by a method described by the user does not create a bank or wallet integration.

3 Feature scope register

The following register is shared by D1, D2 and Project Scope. Must Have means required for the minimum successful FYP; Should Have means intended but deferrable; Could Have means optional if time permits. Priority is not a claim of implementation.

Table 1: Canonical feature scope and priorities

ID	Capability	Priority
SC-01	User registration and authentication	Must Have
SC-02	Basic user profile and settings	Must Have
SC-03	Personal digital expense ledger	Must Have
SC-04	Photograph or upload receipts	Must Have
SC-05	OCR processing	Must Have
SC-06	Structured extraction of merchant and date where available, line items, quantities where available, prices and receipt total	Must Have
SC-07	Review OCR results and edit incorrect extracted values before saving	Must Have
SC-08	Manual quick expense entry	Must Have
SC-09	Text-based cash expense entry	Must Have
SC-10	Voice-based cash expense entry	Should Have
SC-11	Automatic transaction and line-item categorisation	Must Have
SC-12	User confirmation or correction of AI-generated categories	Must Have
SC-13	Locally understandable expense category labels	Must Have
SC-14	Full internal Chart of Accounts	Must Have
SC-15	Mapping of user-facing categories to accounting accounts	Must Have
SC-16	Accounting-compatible transaction storage	Must Have
SC-17	Transaction history	Must Have
SC-18	Search and filtering	Must Have
SC-19	Category-level spending totals	Must Have
SC-20	Basic monthly spending summaries	Must Have

ID	Capability	Priority
SC-21	Item-level price history	Must Have
SC-22	Semantic product classification	Must Have
SC-23	Natural-language financial querying	Must Have
SC-24	AI Financial Assistant	Must Have
SC-25	Dashboard	Must Have
SC-26	Editing and deleting saved expense transactions	Must Have
SC-27	Secure user-data storage	Must Have
SC-28	English and Urdu localisation	Should Have
SC-29	CSV and/or PDF export	Should Have
SC-30	Reasonable support for ordinary Android-class devices	Must Have
SS-03	Custom category labels	Should Have
SS-04	Simple spending visualisations	Should Have
SS-05	Month-to-month comparison views	Should Have
SS-06	Receipt image archive	Should Have
SS-08	User-created categories	Could Have
SS-09	Offline draft expense entry	Could Have
SS-10	Expense reminders	Could Have
SS-11	Exportable household summaries	Could Have

Legacy aliases are retained for traceability: SS-01 refers to SC-10, SS-02 to SC-29, and SS-07 to SC-28. They do not create additional features. Month-to-month comparison views are optional; answering period comparisons through the AI Financial Assistant is core.

4 Architecture boundaries

4.1 Accounting and category boundaries

DailyKhata must maintain a full internal Chart of Accounts with stable account identifiers, account types and mappings from user-facing expense categories. Full means a coherent internal account framework, including the account classes needed for structured recording, not a complete SME accounting interface. Only expense-

management workflows are committed. A local label is separate from its category identity and account mapping.

Table 2: Illustrative local category mappings

User-facing category	Internal accounting account
Ration	Groceries Expense
Transport	Transport Expense
Dining	Dining Expense
Utilities	Utilities Expense
Healthcare	Healthcare Expense
Education	Education Expense
Entertainment	Entertainment Expense
Household Supplies	Household Supplies Expense

These mappings illustrate the approved architecture; account codes and the final hierarchy are later design decisions. For groceries purchased in cash for Rs. 2,500, a double-entry implementation debits Groceries Expense by Rs. 2,500 and credits Cash by Rs. 2,500. Every journal must balance wherever double-entry is used. Users should not enter journals or account codes. No accounting or regulatory certification is claimed.

A mixed receipt retains its receipt total and separately categorised line items. Category totals must not count the same purchase once at transaction level and again at line-item level. Edits and deletion must also update dependent totals and price history consistently. Accounting corrections must preserve balance where journals are used. These are project-team design constraints derived from the need for trustworthy records.

4.2 Semantic product classification

Original receipt text, individual product identity, brand, canonical product type, expense category and accounting classification are separate concepts. Preserve the original text alongside corrected values and inferred classifications. A canonical product type groups related products without replacing their identities.

Table 3: Illustrative semantic product distinctions

Original text and brand	Product identity and type	Category and account
Surf Excel 1kg Brand: Surf Excel	Distinct Surf Excel 1kg product Laundry Detergent	Household Supplies Household Supplies Expense
Bonus 1kg Brand: Bonus	Distinct Bonus 1kg product Laundry Detergent	Household Supplies Household Supplies Expense

Both products can support analysis of Laundry Detergent spending. Exact price history still distinguishes brand, product and pack size. Comparisons require comparable quantities and units; where these are missing, show recorded prices with a limitation instead of claiming equivalent value. Uncertain product matches remain reviewable. Semantic similarity is not proof that two products are identical.

4.3 AI Financial Assistant

The AI Financial Assistant is a core conversational capability grounded in the current user's authorised DailyKhata records. It supports natural-language questions, spending summaries, comparisons between periods and categories, recurring-spending analysis, product-price comparisons, spending patterns, unusually high recorded spending, explanations of change and possible saving opportunities. It may give personalised general financial suggestions based on available historical data.

Illustrative response, not research: restaurant spending of Rs. 8,400 this month compared with Rs. 5,900 last month gives a difference of Rs. 2,500. The assistant may explain that returning to the earlier level would reduce spending by that amount. It must distinguish recorded facts, calculations and suggestions, state the relevant period, and acknowledge missing data. It does not move money, execute payments or control accounts.

Conversational analysis of repeated past expenses is core; automatically generating recurring expenses is future scope. Data-grounded observations about high spending are core; predictive anomaly detection and advanced financial recommendation systems remain future scope. The team may integrate existing models and services instead of training a foundation model.

4.4 Localisation and delivery limits

English and Urdu localisation is Should Have. The architecture should permit translated labels, right-to-left Urdu presentation, local terminology, Urdu-friendly interaction and potential Urdu interaction with the AI Financial Assistant. An initial English interface does not prove bilingual localisation. Urdu support is not guaranteed until implemented and validated. Local category labels remain Must Have independently of translation.

The application is mobile-first and should support ordinary Android-class devices. Receipt OCR targets reasonably clear printed receipts in supported formats, with correction and manual fallback for failures. No universal receipt accuracy or response-time threshold has been validated. Frameworks, model providers, device test specifications and measurable performance targets are later decisions. Offline draft entry is Could Have and does not imply offline OCR or offline assistant services.

5 Feasibility and quality expectations

The delivery unit is a four-student university FYP. A coherent capture, review, save and query workflow takes priority over optional extensions. Existing OCR and language-model components may be integrated. Model choice, hosting, database and mobile

framework are implementation decisions, subject to data handling, cost and test feasibility. No production capacity, regulatory certification, perfect extraction or uptime guarantee is implied.

Secure user-data storage includes authenticated access, appropriate password handling, protected communication, isolation of each user's data, input validation and safe receipt upload handling. Assistant retrieval must respect the same user boundary. Provider credentials must be protected. These are project requirements, not a claim that security has been implemented or certified.

The minimum workflow must permit review of extracted values, correction of categories, saving an expense, finding it again, and editing or deleting it with consistent dependent results. Missing merchant, date or quantity data must not be silently invented. A mixed-category receipt needs a documented aggregation rule. Product matching uncertainty must remain visible enough to correct.

6 Change control

A proposed change records the previous state, new decision, evidence or reason, affected identifiers and documents, decision authority and date. Update the Decision Log first, propagate the decision to every affected document, then repeat the consistency audit. Preserve retired identifiers as aliases or superseded records. Do not silently change Must Have into Should Have to fit a schedule.

Later requirements work may decompose this scope into business, functional and non-functional requirements, trace each requirement to a source and capability identifier, and define measurable acceptance criteria. It must not inherit illustrative hospital examples, concurrency values or performance targets from the D3 teaching brief as DailyKhata requirements. No Deliverable 3 is created here.

7 References

[S1] DailyKhata. Target Market & Audience Analysis. Original submitted D1, dailykhata-d1.docx and corresponding six-page PDF, 01_Deliverable_1. Undated.

[S2] DailyKhata. Original submitted D2, dailykhata-d2.pdf, 02_Deliverable_2, 12 pages. Cover repeats the D1 title; body contains problem identification, stakeholder identification and requirement elicitation. Undated.

[A1] Deliverable 1 task.txt, 01_Deliverable_1/task. User, problem, expectation and target-market assessment criteria. Undated.

[A2] Deliverable 2 task.txt, 02_Deliverable_2/task. Official D2 title and evidence-based assignment requirements. Stated deadline: 25 August 2026, not proof of submission.

[G1] DailyKhata Scope Baseline v1.0 and DailyKhata Documentation Standard v1.0, 00_Project_Governance. Both dated 23 September 2026.

[U1] Project owner instructions for reconstruction, 23 September 2026. Authoritative scope, team identifiers, priorities and constraints.

[A3] Deliverable 3 Requirements Specification, 03_Deliverable_3/task/Deliverable 3.pdf, five pages. Read for compatibility only.

[M1] Design Thinking.pdf, 02_Deliverable_2/task, six instructional slides. Observe, interview, distinguish observation from interpretation, and identify needs.

[M2] Important Links.txt, 02_Deliverable_2/task, 19 teaching-video links. Retained as teaching references; video contents are not treated as research evidence.